

Yash Mehrotra

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[GitHub](#) | [Portfolio](#) | [LinkedIn](#)

Education	
IIT Madras BS Data Science and Application 2023 - Present	• Lead : Linux Community, Google Student Developer Club, IIT Madras. • Core : Webops Team, BS Degree, IIT Madras.
BBD Institute of Technology and Management B.Tech Data Science and applications 2024 - Present	• Relevant Coursework: Data Structures, Algorithms, Database Management Systems, Operating Systems, Computer Networks, Software Engineering.
INTERNSHIP EXPERIENCE	
IIT Madras Data Science Intern Dec-2024 - May-2025	• Built a real-time vehicle detection and counting system using YOLOv5 + SORT, with direction-aware logic and automated CSV reporting for traffic flow analysis. • Designed a secondary detection pipeline using YOLOv8 to identify tire configurations and estimate axle counts, enabling rule-based vehicle classification by axle type without any retraining. • Optimized and deployed YOLOv5/v8 models on NVIDIA Jetson edge devices using TensorRT, significantly reducing inference latency for real-time field deployment. • Engineered a Flask-based surveillance platform supporting live RTSP streaming, configurable tripline event logic, and real-time Telegram alerting for automated incident notification. • Integrated an Annoy-based approximate nearest-neighbor face recognition module for tailgating detection and access control — requiring no model retraining, with embedding-based identity matching at runtime.
SkyGad Ai Development Aug-2024 - Dec-2024	• Engineered a high-performance company website using Next.js and FastAPI, improving load times by 40% and supporting 1000+ concurrent users. • Fine-tuned LLAMA-2 model for mental health applications, achieving 15% improved accuracy and 30% reduction in inappropriate responses. • Developed an advanced RAG pipeline with custom indexing and ranking, reducing query time by 50% and increasing result relevance by 35%.
Projects	
RAVA - Retrieval-Augmented Virtual Assistant	• Made Using: Python, LangGraph, Google Gemini API, Annoy (Approximate Nearest Neighbors), Flask, JSON-based persona & session storage. • Purpose: To build an intelligent, memory-enabled assistant that can retrieve and generate responses using user-specific chat history and custom bot personas. • Benefits: Provides personalized, context-aware responses for each user session, enables fast nearest-neighbor search for relevant information, and supports persistent assistant personalities.
BhashaSarthi	• Made Using: Flask framework and the googletrans library of Python. • Purpose: Bhasha Sarthi is designed to facilitate seamless translation and language understanding, leveraging Google's powerful translation capabilities to break language barriers. • Benefits: It aids in making content accessible to a wider audience by providing accurate translations, thus enhancing communication and understanding across different languages.
Promotify	• Made Using: Flask framework and SQLite3 for database management. • Purpose: Promotify is likely a platform aimed at promoting content or services, utilizing Flask for server-side operations and SQLite3 for storing and managing data efficiently. • Benefits: It helps in organizing promotional campaigns or content in a structured manner, making it easier to reach and engage with the target audience through effective data management and web services.
Chatwallah	• Made Using: Flask, Socket.IO for real-time web socket communication, along with JavaScript, HTML, and CSS for the frontend. • Purpose: Chatwallah is designed as a real-time chat application, facilitating instant messaging and communication among users. • Benefits: It enhances user interaction by providing a platform for real-time conversations, supported by a responsive and user-friendly interface, thanks to the combination of Flask, Socket.IO, and modern web technologies.
Data Analysis on Zomato Dataset	• Made Using: Pandas, Matplotlib, Seaborn. • Purpose: This project aims to analyze the Zomato dataset to uncover insights into customer preferences, popular cuisines, and restaurant ratings. • Benefits: By leveraging data visualization and analysis techniques, it provides valuable insights that can help businesses understand market trends and customer behavior.
Emotion Detection with YOLOv5	• Made Using: Custom trained YOLOv5 model. • Purpose: The project focuses on detecting faces in images or real-time video streams and identifying emotions such as anger, happiness, disgust, and sadness. • Benefits: It can be used in various applications like security systems, customer feedback analysis, and interactive marketing strategies by understanding user emotions.
Co-Curricular Activities	
Public talks and Workshops	• Speaker at GDSC, IIT Madras on the topic "Dumping of Windows", 2024. • Speaker at GDSC, IIT Madras on the topic "Git and GitHub", 2024. • Host and organizer for the workshop "AI Horizons: Navigating the Future of Technology" with "Prof. Sudarshan Iyengar" in 2024, online. • Arranged the workshop "Liftoff" with "Agnikul's Co-Founder" at IIT Madras in 2024, offline.
Achievements	• Runner up in Smart India Hackathon 2023
Skills	
Web Development	NodeJs, ExpressJs, ReactJs, NextJs, FastAPI, SocketIO
Machine Learning	Scikit Learn , Pytorch , Pandas , Numpy , Matplotlib , Seaborn, openCV
Mobile App Dev	React Native
Automation & DevOps	Git, GitHub, Docker
Database management	MongoDB, Redis, SQLite3 , SQL, Postgresql